

4.28)

Mittelwertsatz

$$\begin{aligned} v_m &= \frac{1}{2\pi} \int_0^{2\pi} v(t) dt = \frac{1}{2\pi} \left[\int_0^{\theta} 110\sqrt{2} \sin(t) dt + \int_{2\pi-\theta}^{2\pi} 110\sqrt{2} \sin(t) dt \right] = \\ &= \frac{110\sqrt{2}}{2\pi} \left[-\cos\theta + 1 - \underbrace{\cos(2\pi)}_{=1} + \underbrace{\cos(2\pi-\theta)}_{\cos\theta} \right] = 0 \end{aligned}$$

RMS Mittelwertsatz

$$\begin{aligned} v_{rms}^2 &= \frac{1}{2\pi} \int_0^{2\pi} v^2(t) dt = \frac{1}{2\pi} \left[\int_0^{\theta} (110\sqrt{2})^2 \sin^2(t) dt + \right. \\ &\quad \left. + \int_{2\pi-\theta}^{2\pi} (110\sqrt{2})^2 \sin^2(t) dt \right] = \\ &= \left(\frac{110\sqrt{2}}{2\pi} \right)^2 \left[\int_0^{\theta} \frac{1}{2} (1 - \cos(2t)) dt + \right. \\ &\quad \left. + \int_{2\pi-\theta}^{2\pi} \frac{1}{2} (1 - \cos(2t)) dt \right] = \\ &= \left(\frac{110\sqrt{2}}{2\pi} \right)^2 \frac{1}{2} \left[\left[t - \frac{1}{2} \sin(2t) \right]_0^{\theta} + \left[t - \frac{1}{2} \sin(2t) \right]_{2\pi-\theta}^{2\pi} \right] = \\ &= \left(\frac{110\sqrt{2}}{2\pi} \right)^2 \frac{1}{2} \left[\theta - \frac{1}{2} \sin(2\theta) + 2\pi - 2\pi + \theta - \frac{1}{2} \sin(4\pi) + \right. \\ &\quad \left. + \frac{1}{2} \sin(4\pi - 2\theta) \right] = \\ &= \left(\frac{110\sqrt{2}}{2\pi} \right)^2 \frac{1}{2} (2\theta - \sin(2\theta)) \end{aligned}$$

DWS

$$v_{rms} = 55 \sqrt{\frac{2}{\pi}} (2\theta - \sin(2\theta))^{\frac{1}{2}}$$

Skizze:

$$v_{rms} = 55 \sqrt{\frac{2}{\pi}} (2\theta - \sin(2\theta))^{\frac{1}{2}}$$